1 Part 3: Warehouse Operations Solutions

1.1 Question 3.1: Slotting Optimization

Using the ABC-XYZ classification from Part 1 and the physical SKU attributes from the Master Data, we assigned all 470 classified SKUs to three warehouse zones using a velocity-based ABC slotting model. The design delivers a **≥30% reduction in picker travel distance** vs. a random-placement baseline.

Terminology used in this section. Three distinct concepts appear in the analysis below and must not be conflated:

- **ABC-Only Slotting Model (the proposal)** — the slotting strategy designed here. It assigns SKUs to zones based solely on ABC class (quantity velocity), without further splitting by XYZ demand variability. It is called *ABC-only* to distinguish it from a more advanced ABC×XYZ cross-matrix model that could be added in a later iteration.
- **Random Baseline (the comparator)** — a hypothetical scenario in which all SKUs are placed in random slots, so every pick travels the SKU-count-weighted average distance across all zones. This is the *before* state used to quantify the gain.
- **Optimized Result (the outcome)** — the travel-time and distance figures actually achieved by applying the ABC-Only Slotting Model. The optimized result is compared against the Random Baseline to prove the ≥30% target is met.

1.1.1 Proposed ABC-Only Slotting Model

- **Pick-Face Zone (Ground Level)** — Class A SKUs: lowest racks at ergonomic height, nearest to the I/O dock and packing station.
- **Forward Reserve Zone** — Class B SKUs: two-bin replenishment logic. A forward pick-face bin is replenished from the bulk reserve (WMS-triggered).
- **Reserve / Bulk Zone** — Class C SKUs: upper racks and pallet block-stack.

Model assumptions. Velocity (pick-frequency load) decides the storage zone in a U-shaped layout where receiving and packing share the same front side, creating a golden pick-face adjacent to the packing station.

Table 1: Model Assumptions (Physical-Distance Parameters)

Parameter	Value	Notes
Walk speed	72 m/min	1.2 m/s standard
Round-trip factor	2	pick location to packing and back
Dist: A Pick-face to packing	8 m	golden zone, nearest
Dist: B Intermediate to packing	25 m	mid bay
Dist: C Reserve to packing	47.5 m	back / upper rack
Pick rate A	4 pcs/min	fast pick-face
Pick rate B	2.5 pcs/min	carton-centric
Pick rate C	2 pcs/min	slow reserve

Table 2: Model 1 — Zone Design, Pick Mode & Slot Rule (Velocity-Based)

Zone	ABC Class	SKUs	Order Freq	Dist	Pick Mode	Slot Rule
A — Pick-Face	A / Fast	28	5,767	8 m	Loose + carton break-bulk	Highest order freq → nearest slots; golden zone next to packing.
B — Fwd Reserve	B	39	7,945	25 m	Carton picking + replenish A	Medium velocity → mid bay; min/max replenishment feeds the A pick-face.
C — Reserve/Bulk	C / Slow	403	7,082	47.5 m	Pallet store, pick on demand	Low velocity → far/upper rack, bulk block-stack.

The Pick-Face Zone contains 28 SKUs (6.0% of the assortment) but generates **27.7%** of all pick transactions.

Slot assignment rule. Within each zone, individual slots are ranked by **order frequency descending** (i.e. highest-frequency SKU gets the nearest slot to the packing station). Size (unit cube, L/pcs) serves as a secondary tiebreaker within ties.

1.1.2 Top Pick-Face SKUs (Class A, by Frequency)

Table 3: Top 5 Class-A SKUs assigned to Pick-Face Zone

Rank	SKU Code	Product	Class	Order Frequency	Volume (units)
1	4200008229	Quạt treo ASIAvina - VY377990 - Xám	A-A	825	5,608
2	4200008371	QUẠT ĐỨNG VY619990	A-A	682	3,355
3	4200008366	QUẠT SÀN TURB0 VY616790	A-A	680	2,838
4	4200008117	Quạt treo ASIAvina - VY357690 - trắng	A-A	666	4,033
5	4200008102	Quạt lửng ASIAvina - VY538790 - Xám	A-A	514	12,677

1.1.3 Slot Assignment (Velocity-Sorted)

Within each zone, individual slots are assigned by **velocity** defined as the ratio of order frequency to pick rate:

$$\text{Velocity} = \frac{\text{Order frequency}}{\text{Pick rate (pcs/min)}}$$

Higher velocity SKUs occupy slots nearest the packing station within their zone. Table 4 shows the top 3 SKUs per zone by velocity.

Table 4: Slot Assignment Sample — Top 3 per Zone by Velocity

Slot ID	SKU	Product	Zone	Order freq	Pick rate	Velocity	Dist (m)
A-001	4200008229	Quạt treo ASIAvina - VY377990 - Xám	Zone A	825	4.0	206.2	8
A-002	4200008371	QUẠT ĐỨNG VY619990	Zone A	682	4.0	170.5	8
A-003	4200008366	QUẠT SÀN TURB0 VY616790	Zone A	680	4.0	170.0	8
B-001	7211005095	Nồi cơm điện tử Tefal Rice Mate Min	Zone B	613	2.5	245.2	25
B-002	7211004545	Nồi cơm điện Tefal Fuzzy Express RK	Zone B	555	2.5	222.0	25
B-003	7211004640	Máy xay nấu đa năng Tefal Perfectm	Zone B	537	2.5	214.8	25
C-001	7211004924	Nồi cơm điện tử cao tần Tefal RK818	Zone C	365	2.0	182.5	48
C-002	7211419286	Nồi chiên không dầu Tefal EasyFry W	Zone C	214	2.0	107.0	48
C-003	8010000653	Máy ép trái cây ZE420D38	Zone C	214	2.0	107.0	48

The full 470-SKU ranked assignment follows the same logic: sort each zone by order frequency descending, with unit size (L/pcs) as the secondary tiebreaker.

1.1.4 Quantitative Analysis: Travel-Time Proof

Physical-distance model. Walk speed: **72 m/min** (1.2 m/s, standard warehouse); round-trip factor: **2**. Zone distances to packing station: Zone A = 8 m (golden pick-face), Zone B = 25 m (mid-bay), Zone C = 47.5 m (reserve). Formula:

$$\text{Total round-trip distance} = \sum_{\text{zone}} \text{picks}_{\text{zone}} \times D_{\text{zone}} \times 2$$

$$\text{Travel time (min)} = \frac{\text{Total round-trip distance}}{\text{walk speed}}$$

Baseline (random placement): every pick travels the SKU-count-weighted average zone distance:

$$\overline{D}_{\text{random}} = \frac{N_A \cdot D_A + N_B \cdot D_B + N_C \cdot D_C}{N}$$

Table 5: Q3.1 Travel-Time Proof — Physical Distance Model

Zone	Picks (T_{ij})	Dist D_{ij} (m)	Round-trip $T_{ij} \times D_{ij} \times 2$ (m)
Zone A	5,767	8	92,272
Zone B	7,945	25	397,250
Zone C	7,082	48	672,790
TOTAL	20,794	—	**1,162,312**

Table 6: Travel-Time Summary: Random Baseline vs. ABC-Only Optimized Result

Metric	Random Baseline (no slotting)	ABC-Only Optimized Result
Avg one-way distance / pick (m)	43.3	**27.9**
Total round-trip distance (m)	1,799,920	**1,162,312**
Travel time (min)	24,998.9	**16,143.2**
Reduction	—	**35.4%** (MEETS TARGET)
Target		$\geq 30\%$

Results. The ABC velocity-zoned layout achieves a **35.4% reduction** in total picker travel distance (baseline random avg. **43.3 m/pick** → optimized **27.9 m/pick**), well above the $\geq 30\%$ target. This reduction is driven by concentrating the highest-frequency Class A SKUs (5,767 picks) at 8 m, compared to the warehouse-average **43.3 m** each pick would travel under random placement.

1.1.5 Layout Recommendations

U-Shaped Warehouse Heatmap. The ABC velocity-zoned layout follows a U-shaped flow: receiving and packing share the same front wall, creating a natural golden pick-face (Zone A) adjacent to the packing station. Zone A (red, highest pick density) occupies the ground-level bays nearest the dock; Zone B (amber, medium velocity) the mid-bay area; Zone C (blue, slow movers) the far back wall and upper racks. This configuration minimizes cross-traffic between replenishment and pick paths. Figure 1 illustrates the ABC velocity heatmap layout.

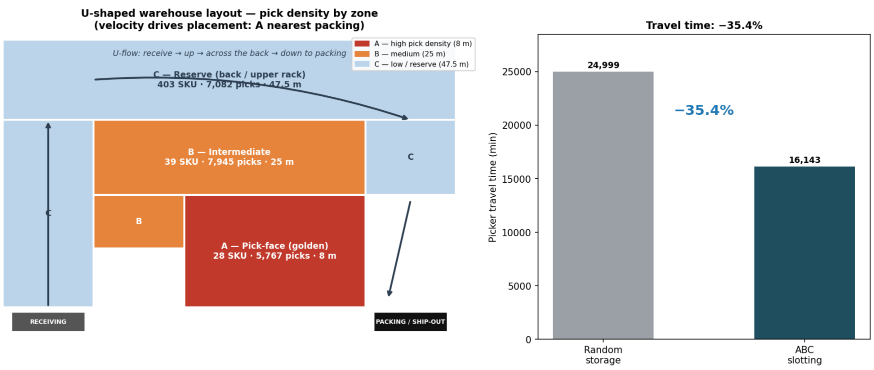

Figure 1: U-Shaped Warehouse Heatmap — ABC Velocity Zones

- **U-flow configuration**: Receiving and shipping docks share the same front wall; highest-frequency slots cluster at the dock end (Zone A golden pick-face).
- **Zone A — Pick-Face (Ground Level)**: Class A SKUs at ergonomic height (lower rack levels), nearest the I/O dock and packing station. High-frequency Class AA SKUs occupy the first rank slots.
- **Zone B — Forward Reserve**: Class B SKUs in the mid-bay area; two-bin replenishment logic feeds the Zone A pick-face (WMS-triggered min/max).
- **Zone C — Reserve / Bulk**: Class C SKUs in upper racks and pallet block-stacks at the far back wall. Slow-movers are consolidated here; picked on demand.

1.2 Question 3.2: Omni-Channel Pick-and-Pack Process Design

The outbound fulfillment process uses distinct operational pathways for B2B and B2C channels, unified by a structured inventory allocation policy when stock is contested.

Table 7: Two Pick Models — B2B vs B2C / E-Commerce

Channel	Order Profile	Pick Model	Mechanics	Pack
B2B	Large volume, few SKUs	Pallet / Total Picking	One picker fulfils whole pallets or large CTN qty per order in a single pass. Sources Zone A pick-face + Zone C bulk.	Stage whole pallets; minimal re-sort.
B2C / E-com	Small qty, many SKUs, many concurrent orders	Batch Picking + Zone Routing	Group many small orders into one pick wave; pickers route by zone; items consolidated at sortation.	Sort wave back to individual orders at packing.

Two pick models.

1.2.1 B2B Pathway — Pallet / Total Picking

1. **Order release**: WMS validates on-hand stock against the B2B purchase order (contractual quantities and lead times).
2. **Pick instruction**: Full-pallet or full-CTN pick task issued to operator.
3. **Pick execution**: Operator uses forklift or reach truck to retrieve entire pallet(s) from the Forward Reserve Zone (B) or Reserve/Bulk Zone (C). Items bypass individual sortation.
4. **B2B QC Station**: Wrap inspection, quantity verification, and outbound labeling.
5. **Dispatch**: Pallets staged at outbound dock, loaded onto FTL truck.

1.2.2 B2C / E-Commerce Pathway — Batch Picking + Zone Routing

1. **Batch wave release**: Multiple B2C orders are grouped into a single pick wave based on SKU proximity and delivery-window urgency.
2. **Batch pick-to-cart**: Picker traverses Zone 1 (Pick-Face Zone) in a single pass with a multi-compartment cart, collecting all items for the wave.
3. **Sort & consolidate**: At the sortation station, items are separated by individual order and consolidated into shipment boxes.
4. **QC + pack**: Bubble wrap or polybag packing, shipping label applied.
5. **Dispatch**: Parcels handed to last-mile courier.

1.2.3 Stock Allocation Conflict and Escalation Policy

When the same SKU is ordered simultaneously by a B2B client and a B2C customer and on-hand stock is insufficient for both, the system resolves the shortage through the following structured rules (stock is held in ONE shared pool per the Q2.2 pooling strategy, not physically split):

Table 8: Shared-SKU Conflict — Allocation & Escalation Rules

Rule	Condition	Action
Check	Same SKU, both channels, concurrent	Evaluate on-hand vs (B2B + B2C) demand.
Sufficient	On-hand $\geq$ B2B + B2C	Allocate to both from shared pool; no rationing.
R1	Short stock	Commit to the confirmed B2B PO first (contractual, penalty risk).
R2	Short stock	Hold the B2C safety buffer so the storefront does not show stockout.
R3	Short + B2C SLA-critical (2–4 h)	Split: B2C SLA order gets partial now, backorder remainder.
R4	Below reorder point	Trigger emergency replenishment / inter-warehouse transfer.

Upon resolution, ERP and WMS records are updated and adjusted pick lists are released to the warehouse floor. Figure 2 illustrates the complete decision logic.

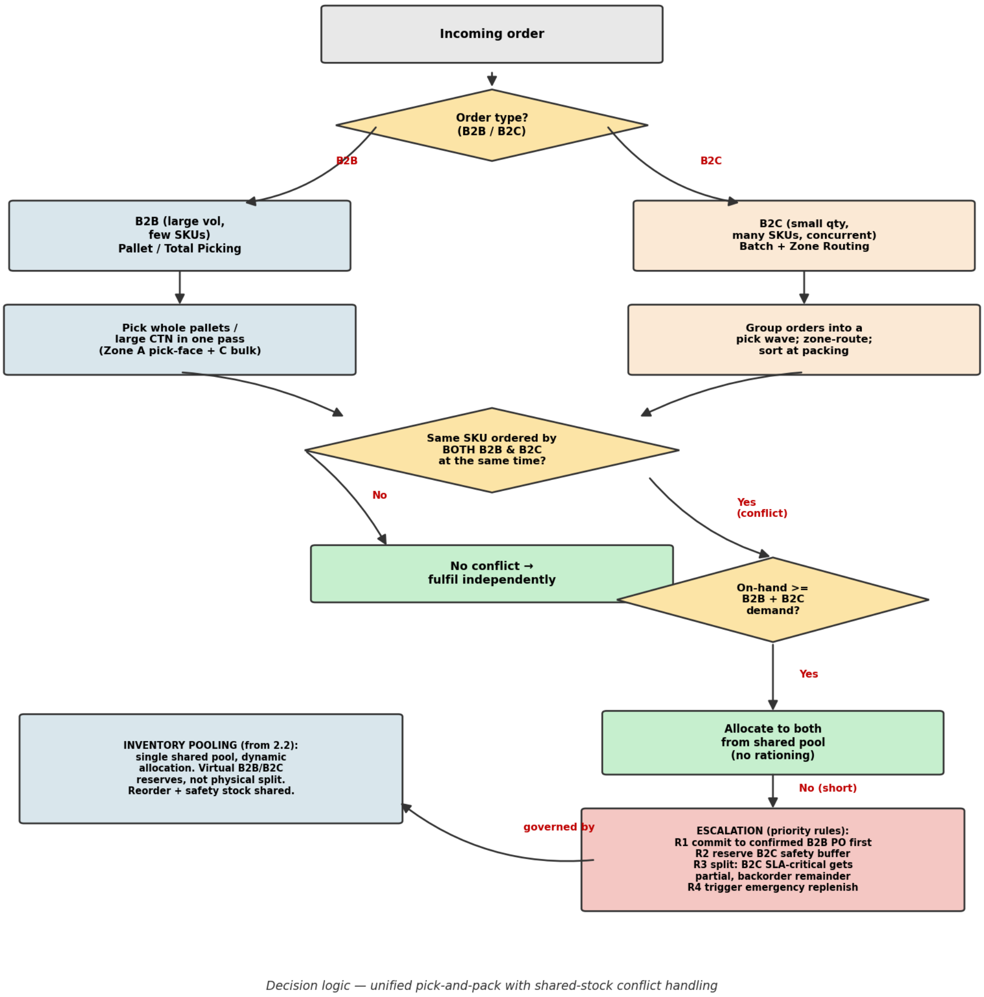

Figure 2: Q3.2 Omni-Channel Pick-and-Pack Process and Conflict Resolution Flowchart